

- **Weighted scores (WS)**: these encompass a range of calculations:
 - Baseline error rate (BER): total errors / total tokens (the “expected” error rate if errors were evenly distributed across features)
 - Feature-specific error rate (FER): errors that bear the feature / total occurrences of the feature in the data set
 - Feature prevalence (PREV): total occurrences of the feature in the data set / total tokens
 - Base weighted score (BWS): $(FER - BER) * PREV$; the subtraction compares “how hard this feature is” relative to the model’s average performance, and the multiplication by prevalence scales that difference by how frequent the feature actually is in the dataset
 - Weighted score rescaled (WSR): $WS * 100$; this scales the score to be readily interpretable
 - Range-normalized score (RNS): $(WSR - \text{MIN}(WSR \text{ for that language-model pair})) / (\text{MAX}(WSR \text{ for that language-model pair}) - \text{MIN}(WSR \text{ for that language-model pair}))$; this places a weighted score in the context of the range of weighted scores for that language-model pair⁶, yielding a 0-1 value

With respect to interpretation of these values,

- The p -value indicates the statistical reliability of an association between feature(s) and error(s).
- The ϕ -value indicates the strength of association between feature(s) and error(s), independent of sample size; the scale used presumes 0-0.29 = weak association, 0.3-0.49 = medium association, 0.5+ = strong association.
- The weighted score (WS) and range of WS’s represent relative difficulty (for the models’ predictive efforts); this value can be positive or negative, wherein a positive value indicates the model struggled *harder* than baseline on a feature, while a negative score indicates that the model did *better* than baseline on a feature. In other words, the WS measures impact relative to baseline, scaled by how frequent the feature is, and is sensitive to both error bias and how frequently the feature occurs (i.e. the practical importance of a feature in the data set).
- The range-normalized score (RNS) makes results cross-comparable across languages/models with very different numeric ranges; the higher a value the worse a model performed and vice versa.
- Taking the The ϕ -value and WS together allows us to evaluate both how strongly a feature is associated with predictive errors (ϕ) and the degree to which it matters in practice (WS).

Chinese

The overall error rates for the two Chinese experiments were: 45.2% of 115 tokens for RFC and 49.6% of 121 tokens for MLP.

Error Type Results

With respect to specific error types, three noteworthy patterns emerged in the results from both classifiers: predicted *le* → actual *suo...de*, predicted *le* → actual *unmarked*, and predicted *le* → actual [adverb] *le*. A further pattern emerged specifically in the RFC experiment: *unmarked* → [varies, all involving *you*], and in the MLP experiment: *suo...de* → [varies].

***Le* → *unmarked*.** This error represents 15.4% of all RFC errors, 25.8% of all *le* overpredictions, and 80% of all *unmarked* underpredictions. Similarly, MLP’s performance showed this error type representing 13.3% of all errors,



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These are over individual features, right? Explicate



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Explain why this is relevant



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But WS is not defined...



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The discussion doesn’t spell out what range of wS reflects. Given my understanding of the measure I’m not sure really...



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What’s baseline?



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Not sure if true...



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Unless you normalize across languages or models (ie have different MIN and MAX functions as defined above)



Le → **unmarked**. This error represents 15.4% of all RFC errors, 25.8% of all *le* overpredictions, and 80% of all *unmarked* underpredictions. Similarly, MLP's performance showed this error type representing 13.3% of all errors, 24.2% of all *le* overpredictions, and 72.7% of all *unmarked* underpredictions. In both models, each of these errors

⁸ For example, a Chinese RFC resultative score of -5.68 is quite good within the -5.68 to 4.31 range of Chinese RFC feature scores, while the German RFC experiential score of -5.67 is rather poor within the -22.90 to -5.67 range of German RFC feature scores.

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was a resultative, and favored achievements at a rate of 6/8 for both models, the remainder being accomplishments.

Le → **suo...de**. This error represents 13.5% of all RFC errors, 22.6% of all *le* overpredictions, and 100% of all *suo...de* underpredictions, and in fact 100% of all instances of actual *suo...de* were *le* predictions. Similarly, MLP's performance showed this error type representing 13.3% percent of all errors, 24.2% of all *le* overpredictions, as well as 88.9% of all *suo...de* underpredictions (MLP correctly predicted *suo...de* once out of nine occurrences in the data set). In both models, each of these errors was a resultative, and favored accomplishments at a rate 5/7 for RFC and 6/8 for MLP, in both cases the remainder were achievements.

Le → **[adverb] le**. This error occurred at a **similar** rate as both the previous error types for both models; however, based on the nature of the error type, it can be argued that this is not a "true" error, as all "mispredictions" did in fact involve actual *le*, only with the addition of a temporal adverbial. For both models, these errors were primarily resultatives with a singular exception being experiential, though as this neatly reflects the distribution of these two features in the data, the relevance of this feature is somewhat lessened. Further, the errors were evenly split between accomplishments and achievements, and all other feature values were 0.

Unmarked → **[varies, incl. you]**. This error represents 11.5% of all RFC errors. In each of these errors the model predicted *unmarked*, and while the actuals varied, each involved the particle *yǒu* (roughly 'have'). The features are mostly generally random in their distribution, though notably there were no resultatives in the set: the tokens were evenly split between experientials and universals.

Suo...de → **[varies]**. This error represents 23.3% of all MLP errors. Of the varying actuals, three were *guo*, five were *le*, and the remainder included *unmarked*, *meiyou...guo*, and *zaoyi*. Most notably about these errors is that each token in this group is an experiential, though the remaining feature values appear random.

The remaining errors in each experiment, comprising 44.2% of RFC errors and 35% of MLP errors, displayed no discernible patterns of significance.

Feature-Value Results

A feature-value analysis on the results of both models revealed two features of potential significance: resultative



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Was this a category or an aggregation over categories? If so, explicate.



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Comparison to featural association of constructions



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Still give numbers



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Plot (or for now: tabulate) these per feature so we can compare ourselves



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What's exp? Spell out!

